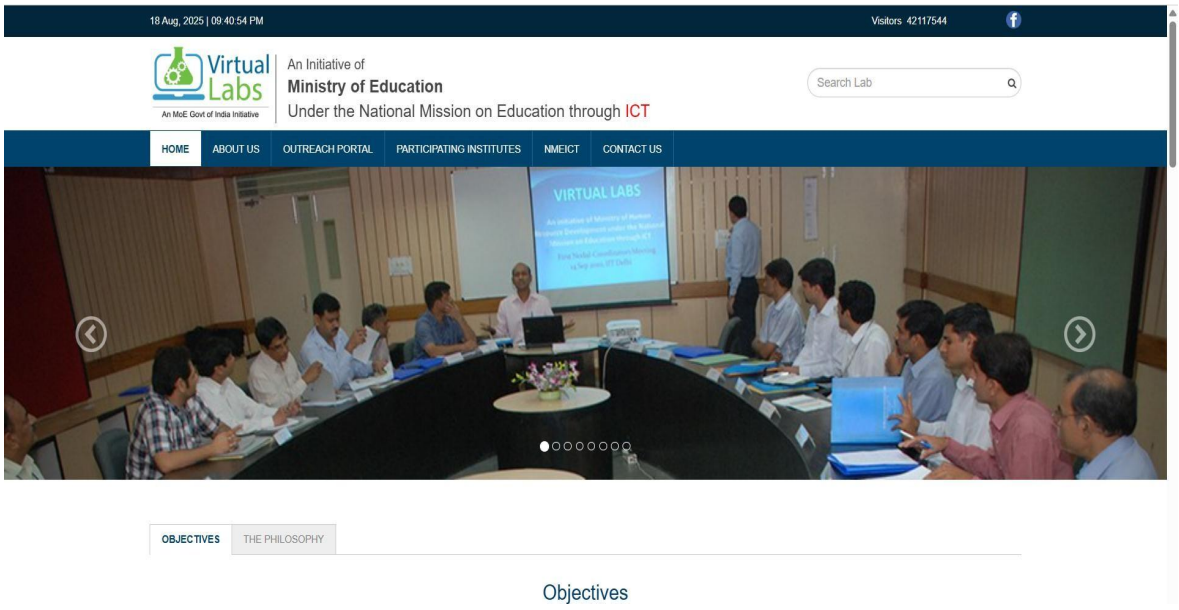
Name of the Student: D.SAIRAM	**Academic Year: 2nd Year**
Student Registration Number: 232P4R2056	**Year & Term: 1st year & 3rd Term**
Study Level: PG	**Class & Section: MCA DS B**
Name of the Course: RPE	**Name of the Instructor:Mahesh prabhu.R**
Name of the Assessment: ASS-2	**Date of Submission: 20-08-25**

Steps to Use Virtual Labs for Experiments

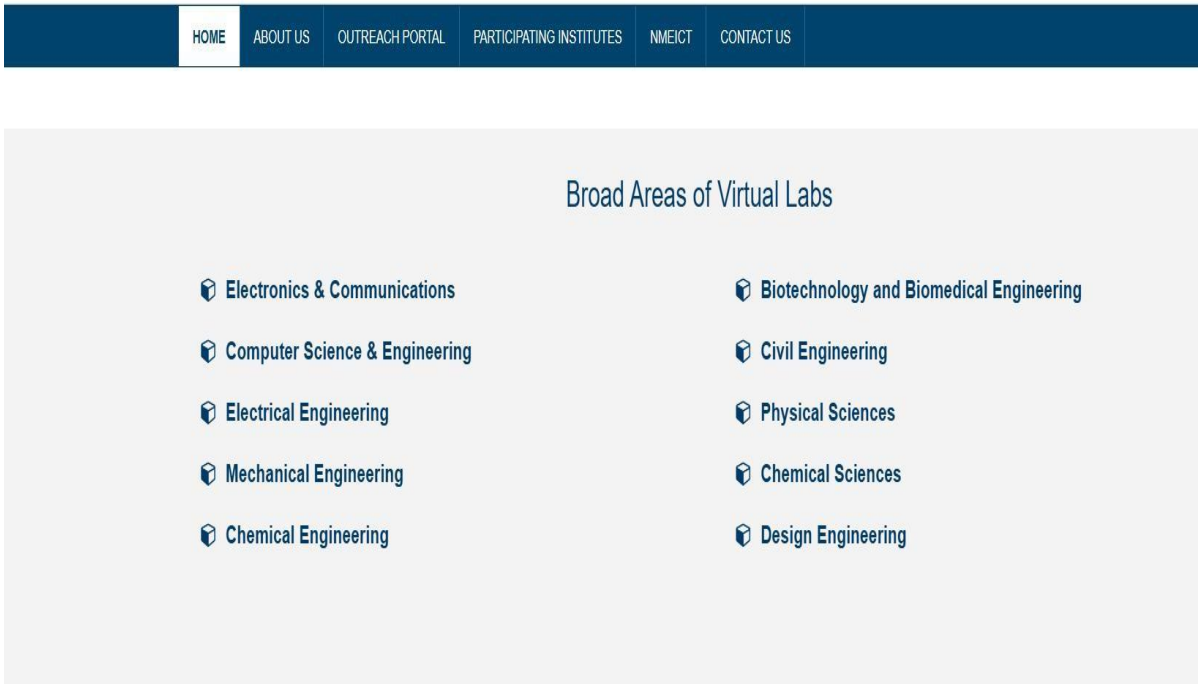
Open the Website

- I began by opening the official Virtual Labs website at https://www.vlab.co.in.



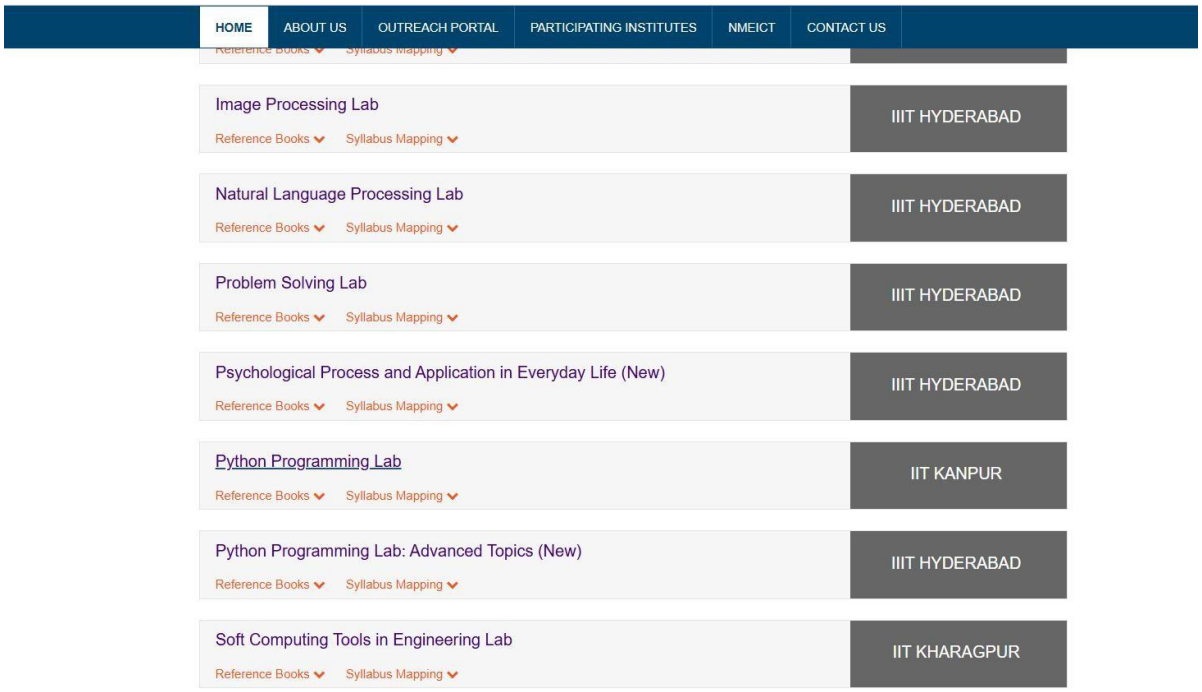
Select Discipline

- Choose a subject area (e.g., Computer Science, Mathematics, Electronics, Physics, etc.).
- From the list of available disciplines, I selected **Computer Science**.



Choose a Lab

- Select a specific lab (e.g., Programming Lab, Data Structures Lab, etc.).
- Within this discipline, I chose the **Python Programming Lab**.

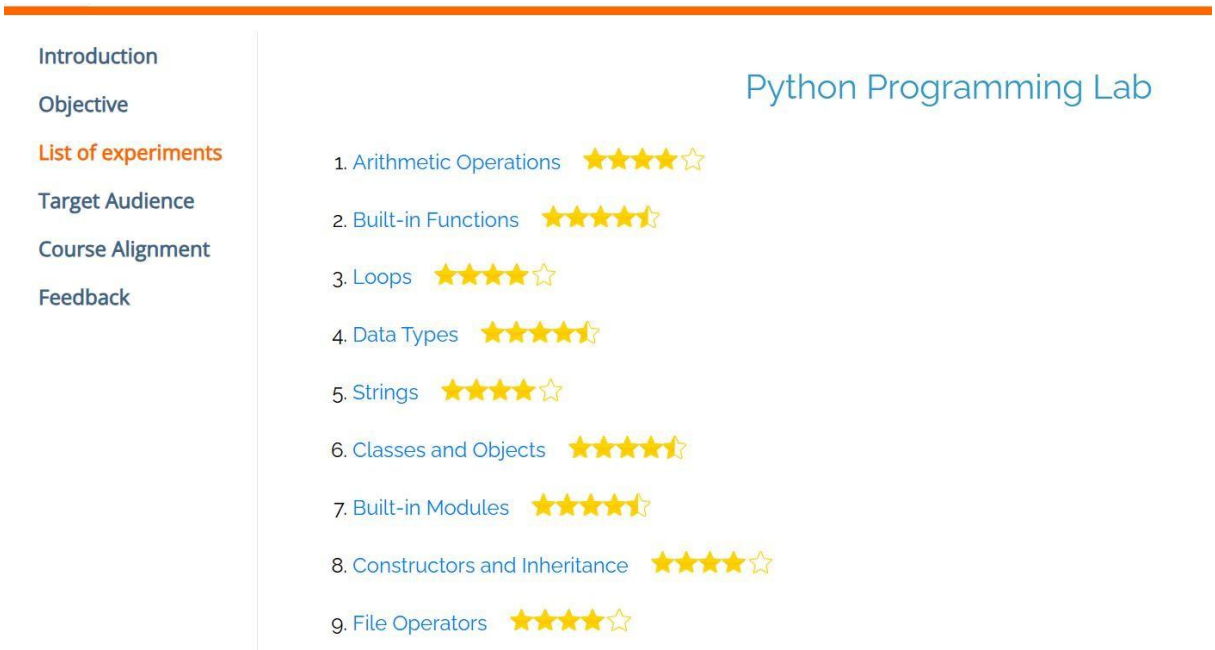


Read the Theory & Procedure

Each experiment page has a **Theory** section (explaining the concept) and a **Procedure** section (explaining the steps).

Pick an Experiment

- Inside the lab, select an experiment (e.g., Arithmetic Operations, Built-in Functions, Loops, Data Types, Strings, Classes and Objects, Built-in Modules, Constructors and Inheritance, File Operators).
- From the list of experiments in this lab, I selected the topic **Loops**.



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- I then clicked on the Simulation tab to run the experiment virtually.

- Using this simulation, I performed experiments such as Factorial Calculation, Fibonacci Series Generation, and Prime Number Checking.

FACTORIAL:

Analysis of Factorial

Fibonacci Series | Prime numbers

Problem?

- To find the Factorial of a Number.

5

Submit

Start | Reset

Step Execute

```
for num in range(start, end + 1):
  factorial = 1
    if num < 0:
      print("Factorial for negative numbers does not exist!!!")
    elif num == 0:
      print("The factorial of the Number is :")
    else:
      for i in range(1,num + 1):
        factorial = factorial*i
      print(The Factorial of the Number is :,num)
```

Result

The Factorial of the Number is :

120

Analysis of Factorial

Fibonacci Series | Prime numbers

Problem?

- To find the Factorial of a Number.

23

Submit

Start | Reset

Step Execute

```
for num in range(start, end + 1):
  factorial = 1
    if num < 0:
      print("Factorial for negative numbers does not exist!!!")
    elif num == 0:
      print("The factorial of the Number is :")
    else:
      for i in range(1,num + 1):
        factorial = factorial*i
      print(The Factorial of the Number is :,num)
```

Result

The Factorial of the Number is :

2.585201673888498e+22

FIBONACCI SERIES:

Analysis of Fibonacci Series

Back

Problem?

- To find the Fibonacci Series.

12

Submit

Start

Reset

Step Execute

```
n1 = 0
n2 = 1
count = 0
if nterms <= 0:
  print("Please enter a positive integer")
elif nterms == 1:
  print("Fibonacci sequence upto",nterms,":")
  print(n1)
else:
  print("Fibonacci sequence upto",nterms,":")
  while count < nterms:
    print(n1,end=' , ')
    nth = n1 + n2
```

Result

Fibonacci sequence upto nterms is :

0,1,1,2,3,5,8,13,21,34,55,89

Loops

Rate Me

Report a Bug

Analysis of Fibonacci Series

Back

Problem?

- To find the Fibonacci Series.

24

Submit

Start

Reset

Step Execute

```
n1 = 0
n2 = 1
count = 0
if nterms <= 0:
  print("Please enter a positive integer")
elif nterms == 1:
  print("Fibonacci sequence upto",nterms,":")
  print(n1)
else:
  print("Fibonacci sequence upto",nterms,":")
  while count < nterms:
```

Result

Fibonacci sequence upto nterms is :

0,1,1,2,3,5,8,13,21,34,55,89,144,233,377,610,987,1597,2584,4181,

PRIME No:

Analysis of Prime no.

Back

Demmonstrate

To find the prime numbers in a Range.

2

6

Start

Next Reset

Step Execute

```
for num in range(start, end + 1):
 if num > 1:
  for i in range(2, int(num/2)):
   if (num % i) == 0:
    break
   else:
    print(" The prime numbers in range {a} and {b} are :".format(a=start,b=end))
    print(num)
```

Result

The prime numbers in range 2 and 6 are :

2 3 5

Analysis of Prime no.

Back

Demmonstrate

To find the prime numbers in a Range.

2

9

Start

Next Reset

Step Execute

```
for num in range(start, end + 1):
 if num > 1:
  for i in range(2, int(num/2)):
   if (num % i) == 0:
    break
   else:
    print(" The prime numbers in range {a} and {b} are :".format(a=start,b=end))
    print(num)
```

Result

The prime numbers in range 2 and 9 are :

2 3 5 7